



Spotify vs 3rd party playlists

Spotify begins cracking down on third party services after scandal reveals debatable 'pay to get featured' issue. <http://dgit.in/Spot3rd>



Hulk eats Cap

In a future comic storyline in the Marvel Universe, a tyrannous Thanos makes an immortal Hulk eat Captain America Steve Rogers. <http://dgit.in/CapOmNom>

how it works will be beneficial to any computer vision job aspirants. It also depends on which job you want.

"It depends almost entirely on which kind of role one is applying for," explains Mr. Ishan Gupta, MD India, Udacity, "(for) someone looking to work on a self-driving car, Simultaneous Localization and Mapping, 3D Object Detection and Tracking, any experience with OpenCV, CUDA, OpenCL and OpenGL would also help. Different applications would have different requirements."

LEARNING

This is where things get a bit tricky. Curriculums in India do not have a singular focus on multiple technology areas, bundling them together into broader, more generic technology courses. Unfortunately, computer vision is one field that is often given this treatment. "There are a very limited number of Indian academic institutions, where a course in Computer Science or Engineering would offer some optional course or elective dealing with Computer Vision," says Mr Ishan, "they might also help students get familiar with Python, digital image processing, and so forth." However, even similar optional courses on areas like Machine Learning or Deep Learning algorithms may not be effective if the quality of such courses varies vastly in different institutions – which is the case. As a result, if you're looking for specialised courses in computer vision, most of the reliable options are online.

A most widely used resources in computer vision is Open Source Computer Vision Library (or OpenCV library). One of the best courses out there to learn more about OpenCV and how it interacts with other computer vision tools is on Udemy at <http://dgit.in/CVUdemy>. You'll also find other courses on the platform that vary in the level of difficulty.

A more comprehensive approach towards mastering computer vision is offered by Udacity's Computer Vision Nanodegree which covers skills like

OpenCV, Computer vision pipeline, neural networks, convolutional neural networks and Keras. Some additional tools that you should learn are TensorFlow, PyTorch and Caffe/Caffe2.



OpenCV is quite popular

THE WORK

Depending on the specific skill set you build and the company you want to work for, the field of computer vision offers quite a few different types of roles. Here's a quick run down of some of these roles and their key responsibilities:

- **Principal Engineer**
 - Creating and reviewing high-level architectural changes for deploying algorithms at scale.
- **Deep Content Indexing**
 - Extracting super metadata from content (Multi modalities).
- **Computer Scientist**
 - Lead and conduct advanced research for Computer Vision.
- **Algorithm Developer**
 - Design, develop, implement and optimise algorithms.
- **Image Processing**
 - Develop auto white balance, auto exposure, auto focus, auto flicker detection/correction, auto scene detection algorithms (e.g)
- **Computer Vision Engineer**
 - Integrate imaging simulation controls, image analysis and data visualization.
 - Can even involve writing code in Python/C++ to implement the algorithms.

So you could be working on making drones fly, or developing surveillance systems that catch criminals red-handed or even building the first truly autonomous car, but if you're working with computer vision, the above roles broadly encompass what kind of tasks you'll be doing on a day to day basis.

Despite all of the advice above, it is true that individuals with hands on experience are given greater importance for these roles. The approach of hiring generalists and training them on the job doesn't work anymore. As Mr Ishan puts it, "With the continuous innovations happening in artificial intelligence and a huge demand for skilled workforce in the area; recruiters are looking for engineers who not only come with theoretical knowledge of these specializations, but also some hands on experience."

IN CONCLUSION

OpenCV was first released in 1999. That should give you an idea about how long we've been at it when it comes to solving the computer vision problem. But today, the capabilities of technology far exceed what we had at our disposal back then and the field is going to explode like never before - with areas like self driving technology, augmented reality, smart security, biometric authentications and more gaining popularity every passing day.

In such a time, you need to master tools like Python, OpenCV, TensorFlow and even Matlab while trying out your own projects to get a strong grip on computer vision. Since specialised courses in educational institutions don't look like a possibility any time soon, online resources are all you've got. That, and your own determination to give computers the faculty of sight.

Resources:

- Udacity Nanodegree: <http://dgit.in/CVUdac>.
- Intro to Computer Vision (Georgia Tech): <http://dgit.in/CVGTech>.
- Advanced Computer Vision: <http://dgit.in/CVAdv>.